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Roll No

EE-8004 (3) (CBGS)
B.E. VIII Semester Examination, June 2020
Choice Based Grading System (CBGS)
Generalized Theory of Electrical Machine
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain Kron primitive machine.
b) With the help of relevant mathematical expression derive the voltage and torque equations for a generalized machine.
2. Discuss the significance of transformations in machine analysis. Explain parkes transformation in detail with the relevant mathematical expression.
3. a) With the help of generalized machine theory, derive voltage equations of a three phase induction machine.
b) Develop the equivalent circuit of a three phase induction motor from generalised concept.
4. a) Based on the concept of generalised theory discuss the modeling of a typical cross field commutation machine.
b) Differentiate between transformer and rotational emf.
5. a) Derive and explain the voltage equations of a synchronous machine in matrix form.
b) Deduce the expressions for various inductances of a three phase synchronous machine.
6. Derive the equations for operational impedances and time constants for a synchronous machine.
7. a) Discuss in detail approximate methods of generator analysis.
b) Discuss the operation of synchronous machine connected to infinite bus bar with the help of phases diagram.
8. Explain the following (any two)
 - a) Approximate methods of power system analysis and its applications.
 - b) Simplified equations of a synchronous machine with two damper coils.
 - c) Schrage motor analysis
 - d) Transient analysis of three phase induction motor when subjected to sudden change in load.
